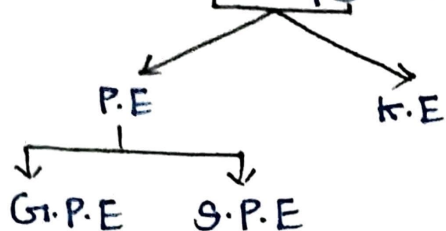


WORK, POWER, ENERGY

Work, Energy, Power



$$W = \vec{F} \cdot \vec{d}$$

$$W = F \times d$$

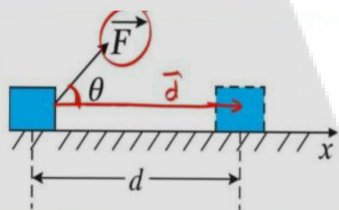


WORK:

The Work done by the force is defined to be the product of components of the force in the direction of the displacement and the magnitude of this displacement.

$$W = (F \cos \theta)(d)$$

Horizontal component of the force. (This component is responsible for displacing the box in horizontal / forward direction).



$$\Rightarrow W = Fd \cos \theta$$

$$\Rightarrow W = \vec{F} \cdot \vec{d}$$

Work is Zero if:

Case - (I) $F = 0$

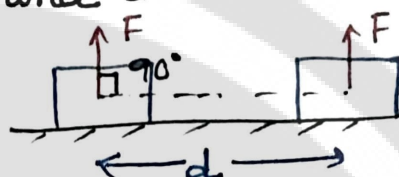
When no force is applied to displace a body from its place then the work done will be zero.

$$W = \vec{F} \cdot \vec{d} = Fd \cos \theta$$

Case - (II) $d = 0$

When the displacement of the body is zero irrespective of the force applied, then the work done will be zero.

Case - (III) When the angle between $\vec{F}$ and $\vec{d}$ is 90° (Perpendicular):



When the force is applied $\perp$ to the displacement of the body, the work done by that force is zero.

$$W = Fd \cos \theta$$

$$W = Fd \cos(90^\circ)$$

$$W = 0$$

Unit of Work:

$$W = \vec{F} \cdot \vec{d}$$

$$(J) (N) (m)$$

$$1 \text{ Joule} = 1 \text{ Nm}$$

SI unit of work

NOTE:

The work done will be max^m when the force is applied in the dirⁿ of the displacement of the body.

Kinetic Energy:

* If an object of mass 'm' has velocity 'v', its kinetic Energy

$$K \text{ is : } K = \frac{1}{2} m v \cdot v = \frac{1}{2} m v^2$$

$$KE = \frac{1}{2} m v^2$$

Scalar velocity (vector)

$\hookrightarrow v^2 \rightarrow$ Scalar

* Kinetic Energy is a Scalar Quantity.

* Measure of the work done by an object by the virtue of its motion is called its kinetic Energy.

Work-Energy Theorem

for a constant force:

WORK ENERGY THEOREM

The change in kinetic energy of a particle is equal to the work done on it by the net force.

$$v^2 - u^2 = 2as$$

Multiplying both sides with $\frac{m}{2}$;

$$\Rightarrow \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2}m \times 2as$$

$$\Rightarrow \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \underbrace{ma}_{\text{displacement}} s$$

$$\Rightarrow \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = Fd$$

$$\underbrace{(\frac{1}{2}mv^2)}_{(K.E)_f} - \underbrace{(\frac{1}{2}mu^2)}_{(K.E)_i}$$

$$\Rightarrow \Delta K.E = (K.E)_f - (K.E)_i = W_{\text{net}}$$

Work Energy Theorem for a constant force.

Work done by a Variable force:

$$W = \vec{F} \cdot \vec{d}$$

$$(e.g. F=10N)$$

constant force.

Variable force:

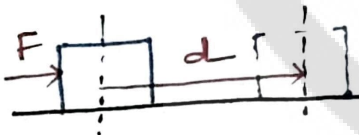
$$F = (2x+1) \text{ N}$$

where x is the displacement initially, $x=0$

$$F = 1 \text{ N} \rightarrow x = 0 \text{ m}$$

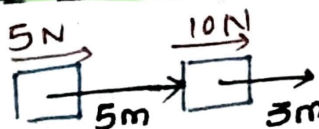
$$F = 3 \text{ N} \rightarrow x = 1 \text{ m}$$

$$F = 5 \text{ N} \rightarrow x = 2 \text{ m}$$

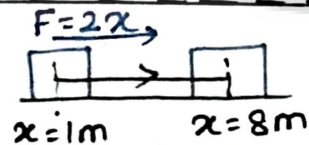


Constant Force Normal Addition

Variable Force Integration



$$W_{\text{net}} = (5 \times 5) + (10 \times 3) = 25 + 30 = 55 \text{ Joule}$$



$$dW = W = \int_{x=1m}^{x=8m} F \cdot dx$$

$$W = \int_{x=1m}^{x=8m} (2x) dx$$

$$W = 2 \int_{x=1m}^{x=8m} x dx = 2 \left[\frac{x^2}{2} \right]_{x=1m}^{x=8m}$$

$$W = [x^2]_{1m}^{8m} = [64 - 1] = 63 \text{ Joule}$$



$$W = \vec{F} \cdot \vec{d}$$

$$\vec{F} \cdot \vec{dx}$$

For a variable force which is a function of displacement, the work done dW for displacing the body to dx displacement is -

$$dW = \vec{F} \cdot \vec{dx}$$

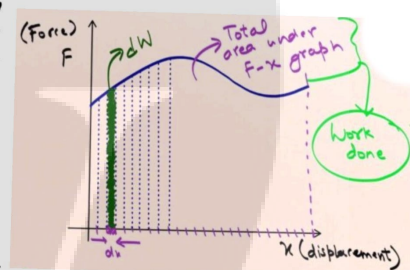
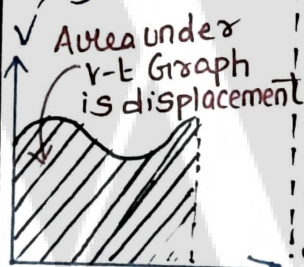
To find total work done, integrate both sides:

$$\int dW = \int F \cdot dx$$

$$\Rightarrow W = \int_{x_i}^{x_f} \vec{F} \cdot \vec{dx}$$

$$\frac{dx}{dt} = v$$

$$\Rightarrow \int dx = \int v dt$$



Graphically work done can be defined as area under the $F-x$ (Force-displacement) Graph.

Work-Energy Theorem for a Variable force:

Since, we know that Work-Energy theorem:

$$W = K_f - K_i$$

So, for variable force, Let's derive this expression by simplifying the rate of change of kinetic energy.

$$\frac{dk}{dt} = \frac{d}{dt} \left(\frac{1}{2} mv^2 \right) = m \frac{d}{dt} \left(\frac{v^2}{2} \right)$$

$$= m \frac{d}{dt} (v dv) = mv \frac{dv}{dt}$$

$$\frac{dk}{dt} = (ma)v \Rightarrow \frac{dk}{dx} = F \frac{dx}{dx}$$

Integrating both sides -

$$\int_{k_i}^{k_f} dk = \int_{x_i}^{x_f} F dx$$

$$k_f - k_i = W$$

Conservative And Non-Conservative Force:

① Conservative Force -

Work done by these forces depend only on the initial and final points of the path and not on the path.

Ex: Gravitational force,
Electrostatic force,
Spring force.

② Non-Conservative Force -

Work done by these forces depend on the path where the force is applied on the body.

Ex: Frictional force,
viscous force.

The Concept of Potential Energy:

Potential Energy (P.E) is the energy possessed by a body due to its position or configuration.

* If a body of mass 'm' is raised to a height 'h' against gravity.

$$U = mgh$$

* Work Energy theorem basis:

Work done against a conservative force gets stored as Potential Energy.

"Jo bhi kaam (work done) humne is ball (mass m) par kiya Ground se Height (h) tak le kar jaane mein, wo workdone Ball mein hi stored rahega in the form of it's Potential Energy."

$$W = \int_{x=0}^{x=h} \vec{F} \cdot d\vec{x}$$

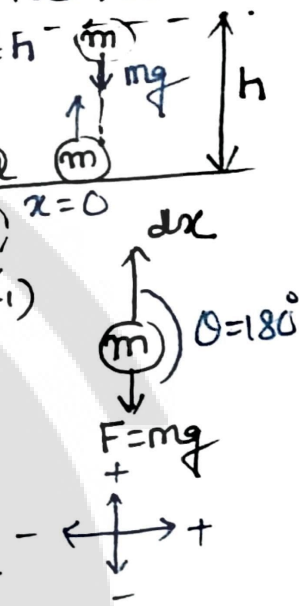
$$W = \int_{x=0}^{x=h} (-mg)(dx) \cos 180^\circ$$

$$W = (-mg)(-1) \int_{x=0}^{x=h} dx (-1)$$

$$W = mg \left[x \right]_{x=0}^{x=h} = mg(h-0)$$

$$P.E \text{ or } U \text{ or } V = mgh$$

Workdone is stored in the form of P.E in the body.



Conservation of Mechanical Energy:

The Principle of conservation of mechanical energy can be stated as:

* The Total Mechanical Energy of a system is conserved if the forces, doing work on it, are conservative.

→ These forces depends only on the initial and final points and not on the path.

→ Total Energy of a system remains constant

$$E_i = E_f$$

$$V_i + K_i = V_f + K_f$$

Derivation Of free fall:

Since, the force acting on the body in free fall is Gravitational force i.e conservative force.

∴ Energy = const

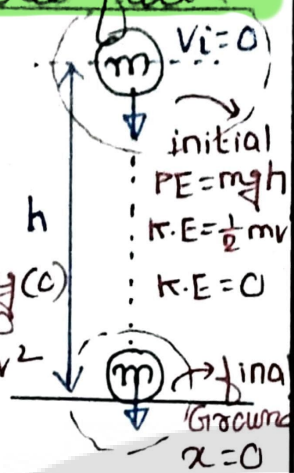
$$\Rightarrow E_i = E_f$$

$$\Rightarrow V_i + K_i = V_f + K_f$$

$$\Rightarrow mgh + 0 = 0 + \frac{1}{2}mv^2$$

$$\Rightarrow v^2 = 2gh$$

$$\Rightarrow v = \sqrt{2gh}$$



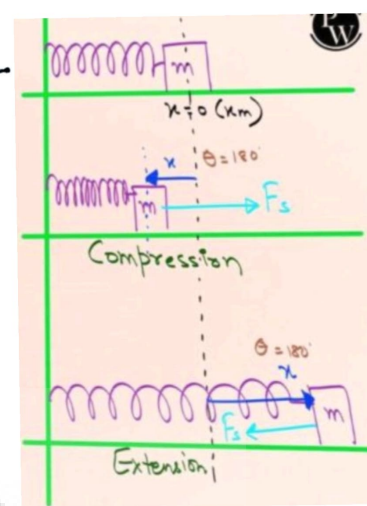
$$W = \int_{x_i}^{x_f} \vec{F} \cdot d\vec{x} \Rightarrow W = \int_{x=0}^{x=x} (-F_s) dx \cos 0$$

$$\Rightarrow W = \int_{x=0}^{x=x} -(-kx)(-1) dx$$

$$= -k \int_{x=0}^{x=x} x dx$$

$$= -k \left[\frac{x^2}{2} \right]_{x=0}^{x=x}$$

$$= -\frac{1}{2} kx^2$$



$$U = -\frac{1}{2} kx^2$$

→ Spring Potential Energy.

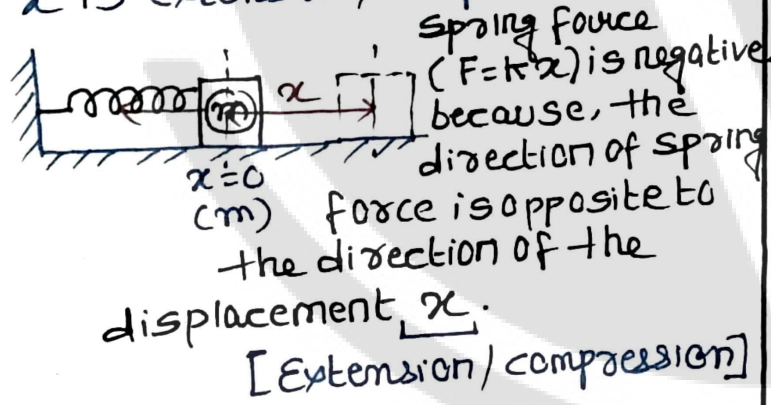
Potential Energy Of a Spring:

HOKE'S LAW:

Restoring Force-

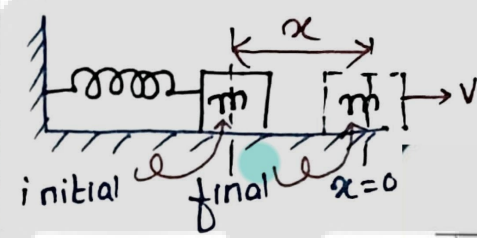
$$F = -kx$$

Where k is the spring constant, x is extension / compression.



Potential Energy of a Spring:

Maximum speed of the Block-



$$E_i = U_i + K_i$$

$$= -\frac{1}{2} kx^2 + 0$$

$$E_f = U_f + K_f$$

$$= -\frac{1}{2} k(0) + \frac{1}{2} mv^2$$

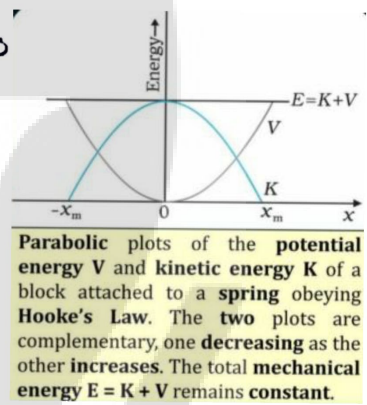
$$= \frac{1}{2} mv^2$$

By conservation of Energy:

$$E_i = E_f$$

$$\Rightarrow \left| -\frac{1}{2} kx^2 \right| = \left| \frac{1}{2} mv^2 \right|$$

$$v = \sqrt{\frac{k}{m}} x$$



Spring Potential Energy:

Work done in compressing or elongating the block of mass m attached to the spring is stored in the block in the form of its spring P.E or Elastic P.E

Power :

Power is defined as the time rate at which work is done or energy is transferred.

Average Power

$$P = \frac{W}{t} \text{ or } P = \frac{E}{t}$$

Instantaneous Power

$$P = \frac{dW}{dt} \Rightarrow P = F \frac{dx}{dt}$$

$$\Rightarrow P = \vec{F} \cdot \vec{v}$$

S.I Unit : Watt (W)

Joule Per Second (J s^{-1})

- There is another unit of Power, namely the horse-Power (hp).
- This unit is still used to describe the output of automobiles, motorcycles etc.
- We encounter the unit Watt when we buy electrical goods such as bulbs, heaters, and refrigerators. A 100 watt bulb which is on for 10 hours uses 1 kilowatt hour (kWh) of energy.

$$1 \text{ HP} = 746 \text{ Watt}$$

$$1 \text{ kW} = 1000 \text{ W} = 10^3 \text{ W}$$

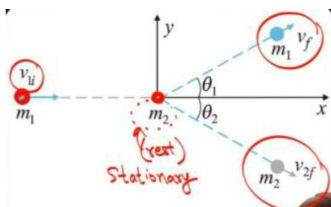
Unit of Work done or Energy

$$1 \text{ kWh} = (1000 \text{ W}) \times (60 \times 60 \text{ Sec})$$

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ W s}$$

Collisions :

* A collision is an event in which two bodies come in direct contact or interact through forces during a very short time interval.



leading to a change in their motion.

* It does not Always mean a physical crash - even interaction through forces (like atomic particles colliding) counts as a collision.

Collisions

Elastic

- one-dimensional collisions.
- Two dimensional collisions

Inelastic

In order to solve collision problems we will use the following:

* Conservation of Momentum
($P_{\text{initial}} = P_{\text{final}}$)

* Conservation of Energy
($E_{\text{initial}} = E_{\text{final}}$)

Elastic & Inelastic Collisions :

collisions are broadly classified into two types :

(A) Elastic collision :

- Both Momentum and Kinetic energy are conserved
- No Loss of energy as heat, sound or deformation.
- Examples: collision between two hard steel balls or gas molecules.

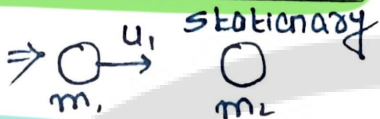
(B) Inelastic collision :

- Momentum is conserved, but kinetic energy is not conserved (some energy is lost as heat, sound or deformation).
- Examples: car crash, clay balls sticking together.

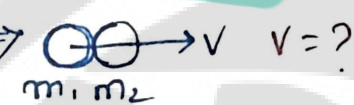
Perfectly inelastic collision:

A special case where colliding bodies stick together after collision and moves as a single body, Here Maximum kinetic Energy is lost.

Collision in One Dimension:

Before collision $\Rightarrow$ 

I Perfectly Inelastic collision -

After collision $\Rightarrow$ 

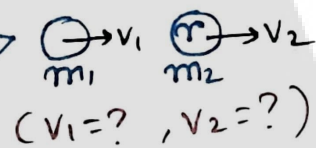
By the Law of conservation of Momentum: $P_i = P_f$

$$\Rightarrow m_1(u_1) + m_2(0) = (m_1 + m_2)v$$

$$\Rightarrow V = \frac{m_1 u_1}{m_1 + m_2}$$

"V" is the velocity with which both the masses move together after collision.

II Elastic collision -

After collision $\Rightarrow$ 

By conservation of Momentum: $P_i = P_f$

$$\Rightarrow m_1(u_1) + m_2(0) = m_1(v_1) + m_2(v_2)$$

$$\Rightarrow m_1 u_1 = m_1 v_1 + m_2 v_2$$

$$\Rightarrow m_1(u_1 - v_1) = m_2 v_2 \quad \text{--- (1)}$$

By conservation of kinetic Energy: $KE_i = KE_f$

$$\Rightarrow \frac{1}{2} m_1 u_1^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$$\Rightarrow m_1 u_1^2 = m_1 v_1^2 + m_2 v_2^2$$

$$\Rightarrow m_1(u_1^2 - v_1^2) = m_2 v_2^2$$

$$\Rightarrow m_1(u_1 - v_1)(u_1 + v_1) = m_2 v_2^2 \quad \text{--- (2)}$$

Now using eqⁿ (1) in eqⁿ (2);

$$\Rightarrow (m_2 v_2)(u_1 + v_1) = m_2 v_2^2$$

$$\Rightarrow u_1 + v_1 = v_2 \quad \text{--- (3)}$$

Put eqⁿ (3) in eqⁿ (1);

$$\Rightarrow m_1(u_1 - v_1) = m_2(u_1 + v_1)$$

$$\Rightarrow m_1 u_1 - m_1 v_1 = m_2 u_1 + m_2 v_1$$

$$\Rightarrow (m_1 - m_2)u_1 = (m_1 + m_2)v_1$$

$$\Rightarrow V_1 = \frac{(m_1 - m_2)u_1}{m_1 + m_2}$$

velocity of m_1 after elastic collision

Put the value of V_1 in eqⁿ (3);

$$\Rightarrow v_2 = u_1 + v_1$$

$$\Rightarrow v_2 = u_1 + \frac{(m_1 - m_2)u_1}{m_1 + m_2}$$

$$\Rightarrow v_2 = \frac{(m_1 + m_2)u_1 + (m_1 - m_2)u_1}{m_1 + m_2}$$

$$\Rightarrow V_2 = \frac{2m_1 u_1}{m_1 + m_2}$$

velocity of m_2 after elastic collision.

Collision in two Dimension:

Before collision: 

After Collision: 

Elastic collision -

$$P_i = P_f$$

x-direction:

$$\Rightarrow m_1 u_1 = m_1 v_1 \cos \theta_1 + m_2 v_2 \cos \theta_2 \quad \text{--- (1)}$$

y-direction:

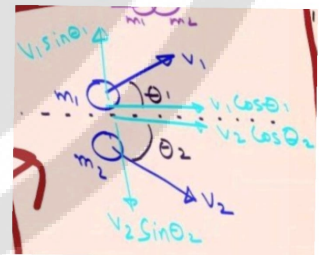
$$\Rightarrow 0 = m_1 v_1 \sin \theta_1 - m_2 v_2 \sin \theta_2$$

$$\Rightarrow m_1 v_1 \sin \theta_1 = m_2 v_2 \sin \theta_2 \quad \text{--- (2)}$$

$$KE_i = KE_f$$

x-direction:

$$\frac{1}{2} m_1 u_1^2 = \frac{1}{2} m_1 (v_1 \cos \theta_1)^2 + \frac{1}{2} m_2 (v_2 \cos \theta_2)^2 \quad \text{--- (3)}$$



y-direction:

$$0 = \frac{1}{2} m_1 (v_1 \sin \theta_1)^2 + \frac{1}{2} m_2 (-v_2 \sin \theta_2)^2$$

$$\Rightarrow m_1 v_1^2 \sin^2 \theta_1 + m_2 v_2^2 \sin^2 \theta_2 = 0 \quad (4)$$

We have 4 variables that we need to find:

$$v_1 = ?$$

$$v_2 = ?$$

$$\theta_1 = ?$$

$$\theta_2 = ?$$

And we have four equations-

$$(1) m_1 u_1 = m_1 v_1 \cos \theta_1 + m_2 v_2 \cos \theta_2$$

$$(2) m_1 v_1 \sin \theta_1 = m_2 v_2 \sin \theta_2$$

$$(3) m_1 u_1^2 = m_1 v_1^2 \cos^2 \theta_1 + m_2 v_2^2 \cos^2 \theta_2$$

$$(4) m_1 v_1^2 \sin^2 \theta_1 + m_2 v_2^2 \sin^2 \theta_2 = 0.$$

Physical Quantity	Symbol	Dimensions	Units	Remarks
Work	W	$[ML^2 T^{-2}]$	J	$W = \mathbf{F} \cdot \mathbf{d}$
Kinetic energy	K	$[ML^2 T^{-2}]$	J	$K = \frac{1}{2} m v^2$
Potential energy	$V(x)$	$[ML^2 T^{-2}]$	J	$F(x) = -\frac{dV(x)}{dx}$
Mechanical energy	E	$[ML^2 T^{-2}]$	J	$E = K + V$
Spring constant	k	$[MT^{-2}]$	Nm^{-1}	$F = -kx$ $V(x) = \frac{1}{2} k x^2$
Power	P	$[ML^2 T^{-3}]$	W	$P = \mathbf{F} \cdot \mathbf{v}$ $P = \frac{dW}{dt}$